COHOMOLOGY OF A TORSION SHEAF, AND OF A COMPLEX WITH TORSION COHOMOLOGY, ON A QUASI-COMPACT QUASI-SEPARATED SCHEME IS TORSION

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file etale-cohomology.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The counted proof is the author's complete argument: the proof is written out in full: the exhaustion of a torsion sheaf by its d-torsion subsheaves with the colimit argument, and the spectral sequence reduction for complexes.

Editorial note. The label lemma-torsion-cohomology names the counted statement, and the proof environment immediately following it in the source is delivered verbatim as its proof.

Editorial note. The counted statement is carried by the author's own lemma environment in the chapter source.

Editorial note. The statement and proof are copied from the pinned chapter file; the two cited inputs are kept as external pointers with their canonical Stacks tags.

Allowed standard prerequisites (editorial).

- Commutation of cohomology with filtered colimits. Same chapter, the lemma computing the cohomology of a filtered colimit of sheaves as the colimit of the cohomologies. Supplies the identity H^n(X, F) = colim H^n(X, F[d]) used in part (1).
- The spectral sequence of a complex computing its cohomology. Derived Categories, the lemma giving the spectral sequence E_2^{p,q} = H^p(X, H^q(K)) converging to H^n(X, K). Supplies the reduction of part (2) to the case of sheaves.

Source reference. Original lemma 015J: sources/stacks-chow/tags.txt, source lines 1510–1510. Prerequisite pointer invoked at proof line(s) 13919; it names a standard fact of a sibling chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Source reference. Original reference 00D5: sources/stacks-etale-cohomology/pinned-original-source.tex, source lines 7427–7427. Prerequisite pointer invoked at proof line(s) 13912; it names a standard fact of the same chapter that the counted argument uses, and every step of the counted proof built on it is retained verbatim in the delivered body.

Lemma 0.1. *Let X be a quasi-compact and quasi-separated scheme.*

(1) *If $\mathcal{F}$ is a torsion abelian sheaf on $X_{étale}$, then $H^n_{étale}(X, \mathcal{F})$ is a torsion abelian group for all n.*
(2) *If K in $D^+(X_{étale})$ has torsion cohomology sheaves, then $H^n_{étale}(X, K)$ is a torsion abelian group for all n.*

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Proof. To prove (1) we write $\mathcal{F} = \bigcup \mathcal{F}[n]$ where $\mathcal{F}[d]$ is the d-torsion subsheaf. By Lemma 00D5 we have $H^n_{étale}(X, \mathcal{F}) = \operatorname{colim} H^n_{étale}(X, \mathcal{F}[d])$. This proves (1) as $H^n_{étale}(X, \mathcal{F}[d])$ is annihilated by d.

To prove (2) we can use the spectral sequence $E_2^{p,q} = H^p_{étale}(X, H^q(K))$ converging to $H^n_{étale}(X, K)$ (Derived Categories, Lemma 015J) and the result for sheaves. $\square$